

# Technical summary: the $G_{19}$ join family

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**Result.** Lalonde's Section 4.2 family is  $G_n = G_{19} \vee K_{n-3}$ ; write it as  $J_n$  to avoid overloading the fixed 19-vertex base graph. The base graph has graph6 string `RxLAKA@AgYAWDGO?0?@??A?W@@OC@_`. For every  $n \geq 3$  and  $d, s \geq 1$ , with  $\xi$  denoting complex orthogonal rank,

$$\boxed{\xi(J_n) = n < \chi_q(J_n) = n + 1}, \quad \chi_q^{[d]}(J_n) = \chi_q^{(s)}(J_n) = n + 1.$$

This confirms Lalonde's candidate finite witnesses for the complex-sphere obstruction and removes the rank-one restriction from his theorem. In particular, for the one-apex graph  $H = J_4$ ,  $\chi_q(H) = 5$ . Zero projectors, nonuniform original ranks, arbitrary finite dimension, and noncommuting projectors are all allowed.

**Uniformization and clique saturation.** Suppose  $J_n$  has a quantum  $n$ -coloring in dimension  $d$ . Direct-sum over all color permutations:

$$\tilde{P}_{v,c} = \bigoplus_{\pi \in S_n} P_{v,\pi(c)}.$$

The new dimension is  $D = n!d = nr$ , and every projector has common positive rank  $r = (n-1)!d$ . Thus uniformity is a theorem-level reduction, not an ansatz. For an  $n$ -clique, each fixed-color column consists of pairwise orthogonal projections. Its trace is at most  $D$ ; summing over all colors and using the row partitions makes all  $n$  inequalities equalities. Hence every clique column sums to  $I_D$ .

Fix a color  $c$ . Let  $R_c$  be the sum of its projections on the joined clique and  $Q_c = I - R_c$ . Each of 123, 189, 345, and 267, joined with  $K_{n-3}$ , is an  $n$ -clique. Therefore

$$\dim Q_c = 3r, \quad \sum_{v \in T_j} P_{v,c} = Q_c.$$

On  $Q_c$ , write  $e_v = P_{v,c}|_{Q_c}$ . These are rank- $r$  projections giving a projective orthogonal representation of  $G_{19}$ .

**Exact core SOS.** Set  $X = e_{10} + e_{11} + e_{12} + e_{13}$ ,

$$\begin{aligned} H_1 &= e_{10} + e_{11} - e_{12} - e_{13}, \\ H_2 &= e_{10} - e_{11} + e_{12} - e_{13}, & (d_1, d_2, d_3) &= (e_5 - e_4, e_7 - e_6, e_9 - e_8). \\ H_3 &= e_{10} - e_{11} - e_{12} + e_{13}, \end{aligned}$$

Projection, edge, and triangle relations give

$$\sum_i d_i^2 = 2I, \quad X^2 + \sum_i H_i^2 = 4X, \quad \sum_i \{d_i, H_i\} = -4X.$$

The final identity is graph-specific: for each  $j = 10, \dots, 13$ , the positive Walsh signs select exactly the three neighbors of  $j$  among  $4, \dots, 9$ . Expanding the following factors now gives the exact rational identity

$$\frac{4}{3}I - X = \sum_{j=0}^3 f_j^* f_j, \quad f_0 = \frac{1}{2} \left( X - \frac{4}{3}I \right), \quad f_i = \frac{2}{3}d_i + \frac{1}{2}H_i.$$

Both sides have trace zero, since  $\dim Q_c = 3r$  and each  $e_j$  has rank  $r$ . Faithfulness of finite matrix trace forces every  $f_j = 0$ .

Put  $A = e_4 - e_5$ ,  $B = e_6 - e_7$ , and  $C = e_8 - e_9$ . Walsh inversion and  $e_j^2 = e_j$  yield

$$\{B, C\} = A, \quad \{A, C\} = B, \quad \{A, B\} = C.$$

Relative to  $Q_c = \text{ran } e_1 \oplus \text{ran } e_2 \oplus \text{ran } e_3$ , support restrictions put  $A, B, C$  on coordinate pairs 12, 13, 23. The anticommutator equations kill their diagonal blocks; the remaining blocks are unitaries with one cocycle relation. A block-diagonal gauge makes all three blocks  $I_r$ . Thus vertices  $1, \dots, 13$  are exactly the scalar sign-vector core tensored with a multiplicity space  $K_c \cong \mathbb{C}^r$ .

**Exhaustive tail.** The remaining ranges are encoded by an arbitrary  $r$ -plane  $M_c \subset K_c \oplus K_c$  invariant under  $J_c = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ :

$$\begin{array}{c|ccc} v & 14 & 15 & 16 \\ \hline \text{ran } e_v & (a, 0, b) & (0, a, b) & (a, -b, 0) \end{array} \quad (a, b) \in M_c,$$

$$\begin{array}{c|ccc} v & 17 & 18 & 19 \\ \hline \text{ran } e_v & (x, x, z) & (x, -z, z) & (x, -z, x) \end{array} \quad (x, z) \in M_c^\perp.$$

The six core–tail edges supply the displayed ambient frames. Pulling back the six tail–tail edges gives five identity maps and one  $-J_c$ , proving both exhaustiveness and  $J_c M_c = M_c$ . This formulation includes non-graph planes;  $M_c = \text{ran}(I, S_c)$  with  $S_c^2 = -I$  is only a transverse chart. At  $r = 1$ ,  $M_c = \text{span}(1, \sigma i)$  and the six displayed frames recover Lalonde’s Theorem 4.5 normal form with sign  $b = \sigma$ .

**Cross-color sector packing.** Choose a core unitary  $W_c : K_c^3 \rightarrow Q_c$ . For  $c \neq d$ , same-vertex orthogonality at vertices  $1, \dots, 9$  forces

$$W_d^* W_c = \begin{pmatrix} 0 & X & Y \\ -X & 0 & Z \\ -Y & -Z & 0 \end{pmatrix}.$$

The lower blocks are negatives, not adjoints, because the matrix maps the  $c$  coordinate space to the  $d$  coordinate space. With  $\Omega_L = \begin{pmatrix} 0 & L \\ -L & 0 \end{pmatrix}$ , the six tail compressions are

$$\Omega_Y, \Omega_Z, -\Omega_X, \Omega_{Y+Z}, \Omega_{Y-X}, \Omega_{Z-X}.$$

Their invertible coefficient transform implies, for  $L = X, Y, Z$ ,

$$\Omega_L M_c \subseteq M_d^\perp, \quad \Omega_L M_c^\perp \subseteq M_d.$$

Since  $M_c$  reduces  $J_c$ , write

$$M_c \cap \ker(J_c - \sigma i I) = \{(x, \sigma i x) : x \in E_{c,\sigma}\}, \quad e_{c,+} + e_{c,-} = r.$$

Each  $\Omega_L$  intertwines  $J_c$  and  $J_d$ , so the plane flip says  $LE_{c,\sigma} \subseteq E_{d,\sigma}^\perp$ . For fixed  $\sigma$ , the  $n$  physical spaces

$$W_c(E_{c,\sigma} \oplus E_{c,\sigma} \oplus E_{c,\sigma})$$

are pairwise orthogonal in dimension  $nr$ . Hence

$$3 \sum_c e_{c,+} \leq nr, \quad 3 \sum_c e_{c,-} \leq nr.$$

Adding gives  $3nr \leq 2nr$ , impossible for  $r > 0$ . Thus no quantum  $n$ -coloring exists. The explicit classical four-coloring of  $G_{19}$  and fresh colors on the joined clique give the matching  $(n+1)$  upper bound.

**Verification.** Two standard-library exact obstruction verifiers and a separate graph checker accompany the proof. They decode the graph6 checksum, verify the published four-coloring and exhaustive non-three-colorability, replay independent rational noncommutative SOS expansions, check the Walsh and skew-block systems, multiply all six tail compressions, invert their coefficient transform, and check the symbolic all- $n$  dimension gap. No floating-point computation enters the theorem.

**Scope and relation to prior work.** The known rank-one theorem motivates the sign frame, but it is not used as a reduction: the SOS proves the module-valued core directly, and the invariant plane analysis handles genuinely higher-rank tails. The contradiction excludes every finite-dimensional representation of the  $n$ -coloring relations, reducible or irreducible. No finite local dimension supports  $n$  colors. At  $n+1$  colors, the classical solution lifts to fixed dimension  $d$  by tensoring with  $I_d$  and to fixed rank  $s$  by taking the direct sum of its cyclic color relabelings and tensoring with  $I_s$ . No claim is made about infinite-dimensional or commuting-operator colorings.